

Describing Data Graphically with R

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Describing Data



Methods in Psychological Research

Hans Rosling

Hans Rosling's 200 Countries, 200 Years, 4 Minutes - Th...



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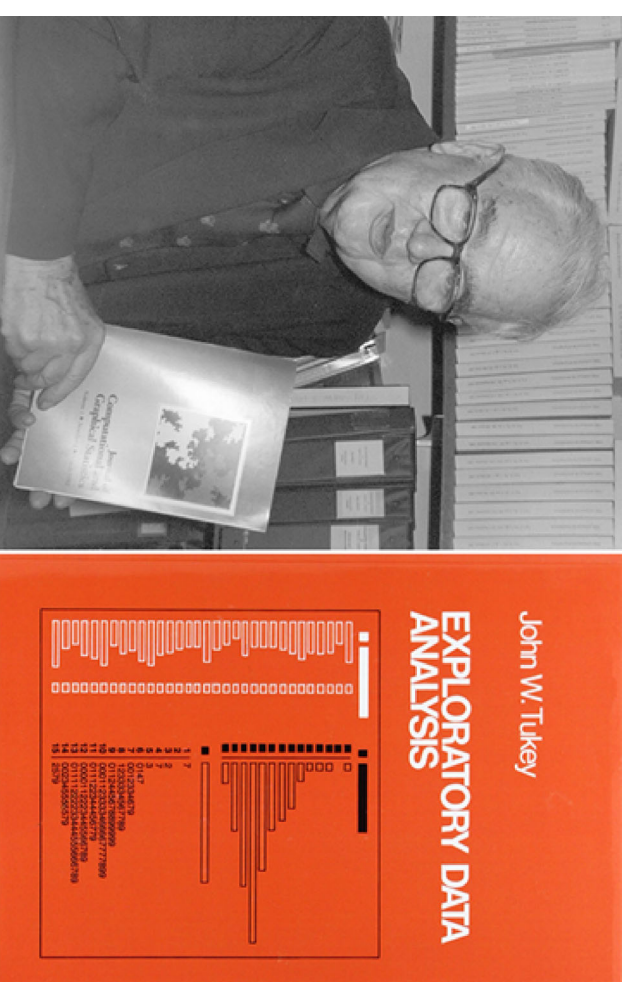
Summarize

- + Transform a pile of numbers into a summary
- + Descriptive Statistics
- + Distribution of a variable is a table/graph showing the categories/values of outcomes and their frequency/percentage of occurrence
- + Exploratory Data Analysis (Tukey, 1977)



Exploratory Data Analysis

- + Tukey (1977)
- + EDA
- + Graphical Data Analysis
- + Numbers as summaries
- + Emphasized Robust Statistics



Descriptive Statistics

- + Examples
- + Tables
- + Graphs
- + Summary Statistics



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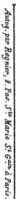
Tables

+ Woodbridge (1845)

POPULATION OF GRAND DIVISIONS.					
	Hassel.	Malte Brun.	Balbi.	Weimar Almanac, 1840.	McCulloch, 1829.
EUROPE, <i>Inhabitants</i> ,	227,000,000	205,000,000	227,700,000	233,240,043	232,677,909
ASIA,.....	470,000,000	340,000,000	390,000,000	608,516,019	401,230,000
AFRICA,.....	110,000,000	104,000,000	60,000,000	101,498,411	
AMERICA,.....	43,000,000	45,000,000	39,000,000	48,007,150	47,000,000
OCEANICA,.....	2,600,000	20,000,000	20,300,000	1,838,194	
Total,.....	852,600,000	714,000 000	737,000,000	993,108,507	



+ Minard (1869)



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Examples

- + Summary Statistics
- + Measures of Central Tendency
- + Measures of Spread



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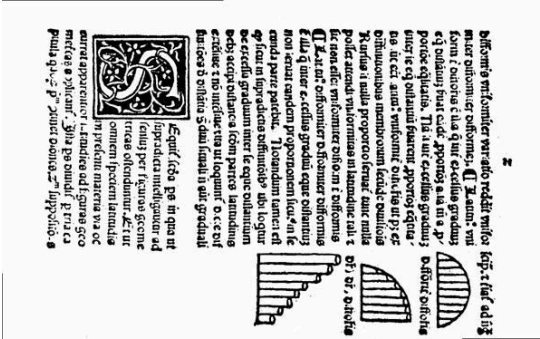
Categorical Variable Displays (Nominal, Ordinal)

- + Frequency Distribution Graphs
 - + Bar Chart (appropriate for nominal and ordinal data)
 - + Pie Chart (best for nominal data with few categories)
- + Quantitative Variables
 - + Histograms
 - + Stem plots
- + Time Plots
 - + (can be used for any level, often interval or ratio)



Frequency distribution graph

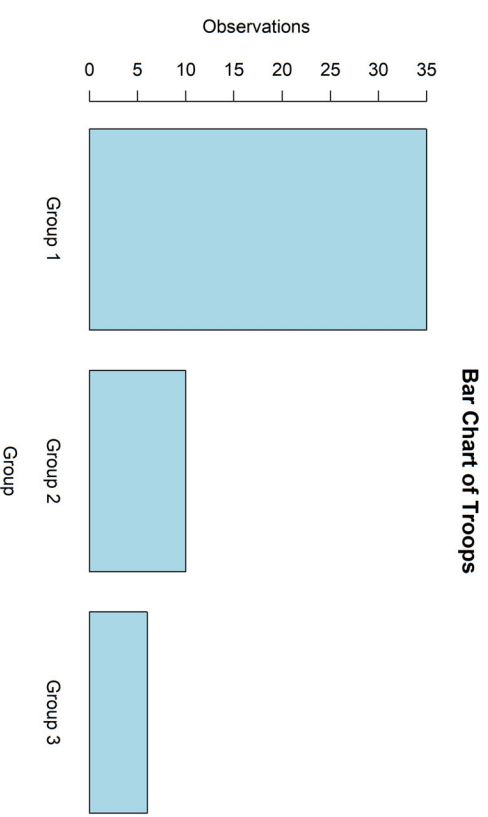
- + Bar Chart
- + Graphs of variables with categories of outcomes on the x axis; and the frequency or percent of each category on the Y axis.
- + Appropriate for nominal and ordinal data



Bar Graph/Chart

```
# Bar chart  
library(HistData)  
Minard_troops_demo <- Minard.troops  
  
Minard_troops_demo$group <- paste("Group", Minard_troops_demo$group)  
counts <- table(Minard_troops_demo$group)
```

```
barplot(counts, main="Bar Chart of Troops",  
        xlab="Group", ylab="Observations",  
        col="lightblue")
```



- + Note: This example uses nominal data (Troop Group) and how many data points we have for each group.
- + Bar charts are suitable for nominal and ordinal data as they show frequency for discrete categories.

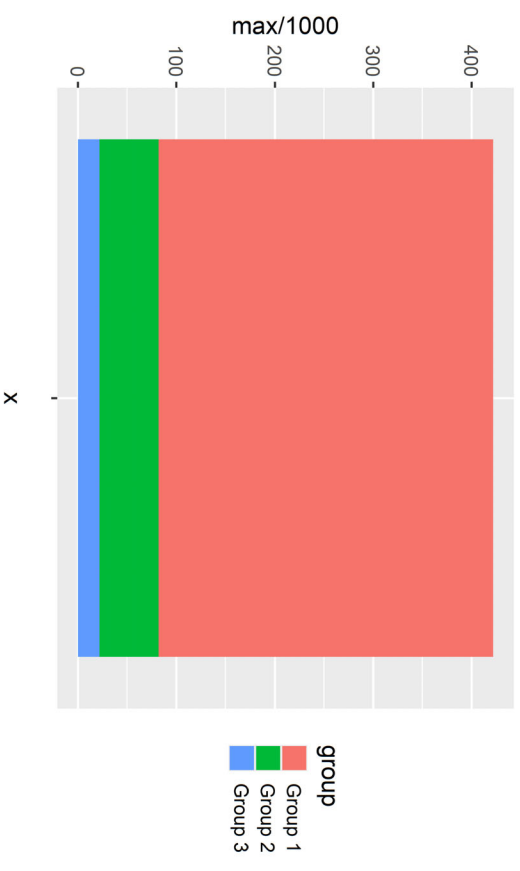


Stacked Bar Chart (Total Troops)

```
df <- data.frame(
  group = c("Group 1",
            "Group 2",
            "Group 3"),
  max = c(340000, 60000, 22000),
  min = c(4000, 28000, 6000))
head(df)
```

```
##      group  max  min
## 1 Group 1 340000 4000
## 2 Group 2  60000 28000
## 3 Group 3  22000  6000
```

```
library(ggplot2)
bp <- ggplot(df, aes(x="",
                     y=max/1000, # to rescale
                     fill = group)) +
  geom_bar(width = 1, stat = "identity")
bp
```

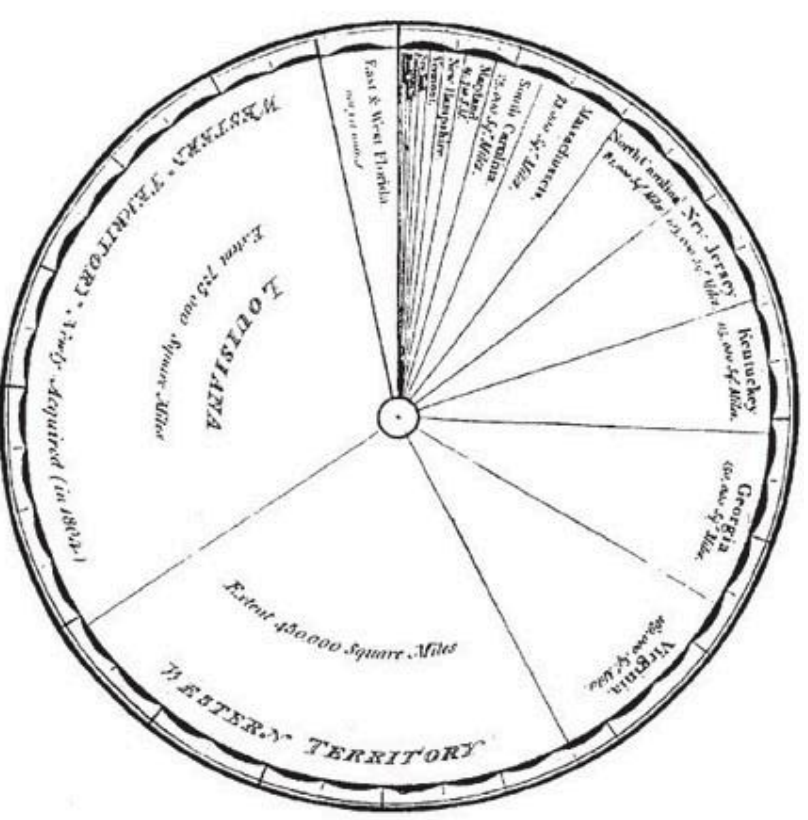


- + Stacked bar charts can show the relationship between two categorical variables.



Pie Chart

- + Graphs of variables with categories of outcomes as frequency or percent of each category in the pie.
- + Best for nominal data with few categories



"A pie chart showing each state in the United States part of it. A playful translation of A Statistical Account of the United States of America by D. F. Donnan. The chart shows the distribution of land area in 1800." The chart is a playful translation of the original, showing the distribution of land area in 1800.

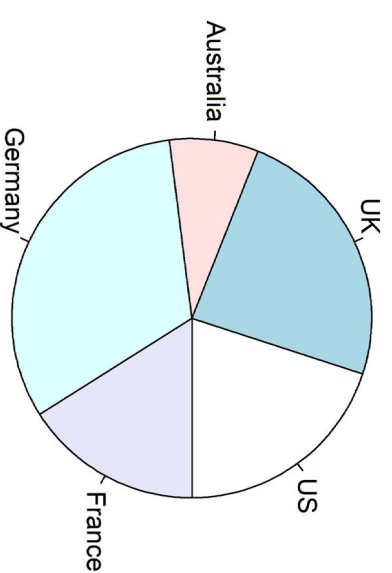


Pie chart

```
slices <- c(10,12,  
            4,16, 8)  
lbls <- c("US", "UK",  
          "Australia",  
          "Germany",  
          "France")
```

```
pie(slices, labels = lbls,  
    main="Pie Chart of Countries")
```

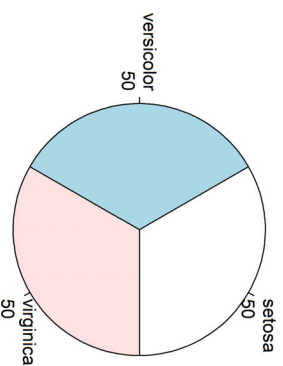
Pie Chart of Countries



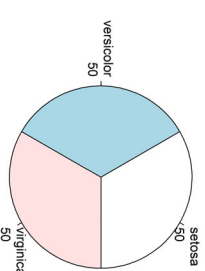
Example 2

```
mytable <- table(iris$Species)
lbls <- paste(names(mytable), "\n", mytable, sep="")
pie(mytable, labels = lbls,
    main="Pie Chart of Species\n (with sample sizes)")
```

**Pie Chart of Species
(with sample sizes)**



**Pie Chart of Species
(with sample sizes)**

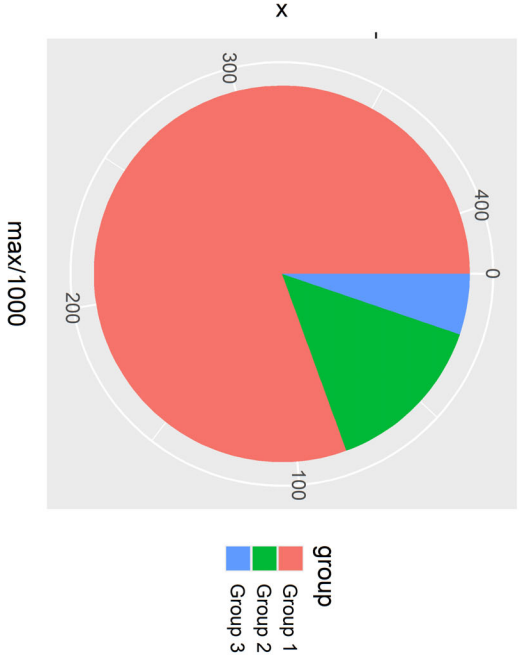


- + Pie charts are best for nominal data with few categories, showing part-to-whole relationships.

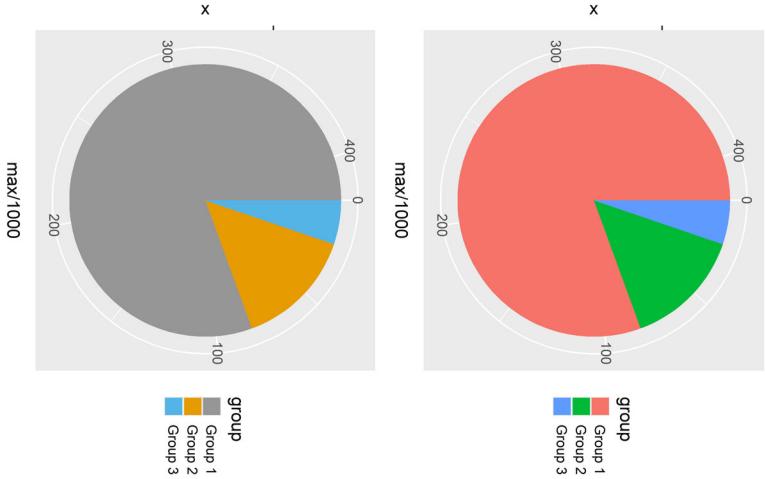
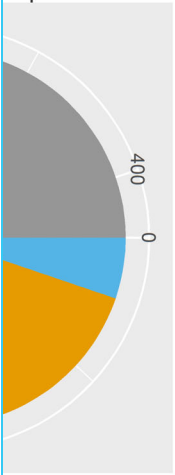


Convert Bar Chart into Pie Chart

```
pie <- bp + coord_polar("y", start=0)
pie
```



```
pie + scale_fill_manual(values=c("#999999", "#E69F00", "#56B4E9"))
```



Quantitative Variables

- + Interval or Ratio Scales
 - + Histograms
 - + Stem plots
 - + Time plots



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Histogram

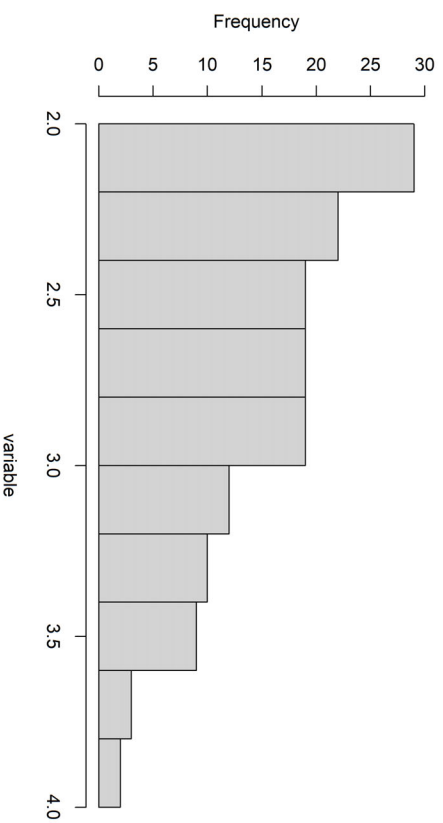
- + A histogram is a graphical representation of the distribution of numerical data.
- + Approximates a probability distribution
- + First described by Pearson in 1895.



Histogram

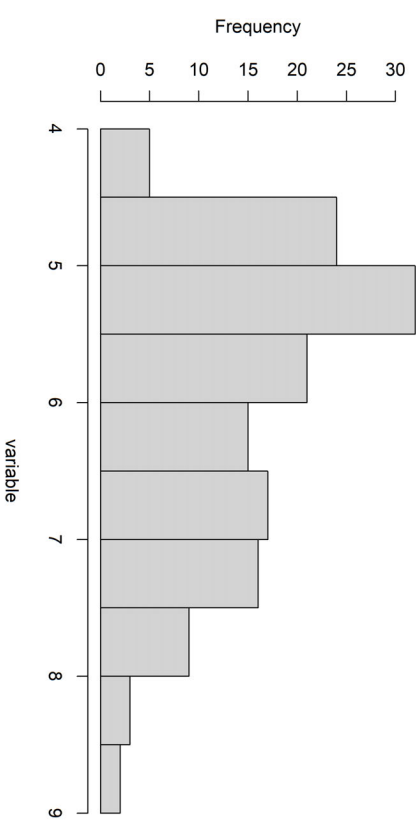
```
library(MASS) # load library
variable<-cats$Bwt
hist(variable)
```

Histogram of variable



```
#Convert to Imperial
variable<-variable*2.2
hist(variable)
```

Histogram of variable



- + Histograms are appropriate for interval and ratio data, showing the distribution of continuous variables.

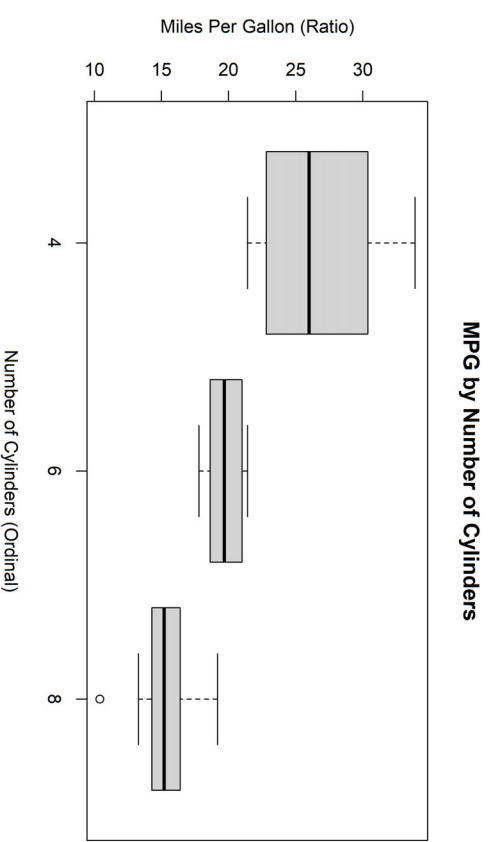


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Boxplot (Ordinal predictor, Ratio outcome)

- + Boxplots can show the distribution of a ratio variable across categories of an ordinal variable.

```
boxplot(mpg ~ cyl, data=mtcars,  
        main="MPG by Number of Cylinders",  
        xlab="Number of Cylinders (Ordinal)",  
        ylab="Miles Per Gallon (Ratio)")
```



Stemplot

- + Sometimes called a stem and leaf diagram
- + A way to display data that splits the data into a stem and leaf.
- + The stem is the first digit of the number and the leaf is the second digit
- + Stem and leaf plots are useful for displaying the distribution of interval or ratio data, especially for smaller datasets.

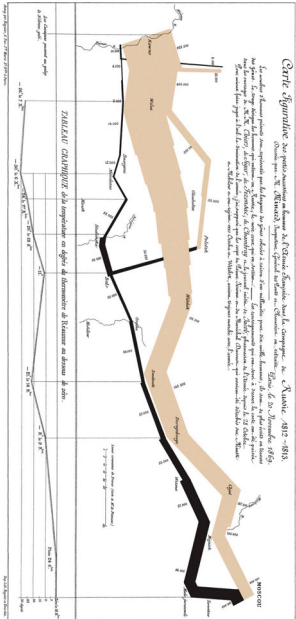
- + Eruption duration is a ratio variable
- + Each leaf represents the ones digit
- + Each stem represents the tens digit

```
# Stem and Leaf plot
stem(faithful$eruptions, scale=1)
```

```
## The decimal point is 1 digit(s) to the left of the |
##
## 16 | 070355555588
## 18 | 000022233333355777777888822335777888
## 20 | 00002223378800035778
## 22 | 0002335578023578
## 24 | 00228
## 26 | 23
## 28 | 080
## 30 | 7
## 32 | 2337
## 34 | 250077
## 36 | 0000823577
## 38 | 233335582225577
## 40 | 000000335778888002233555577778
## 42 | 0333555778800233333555577778
## 44 | 022223355778000000023333357778888
## 46 | 000023357700000023578
## 48 | 00000022335800333
## 50 | 0370
##
```



Time Plots



Edward Tufte has said that Minard's plot:

"may well be the best statistical graphic ever drawn"

It packs a ton of information into one dense figure.

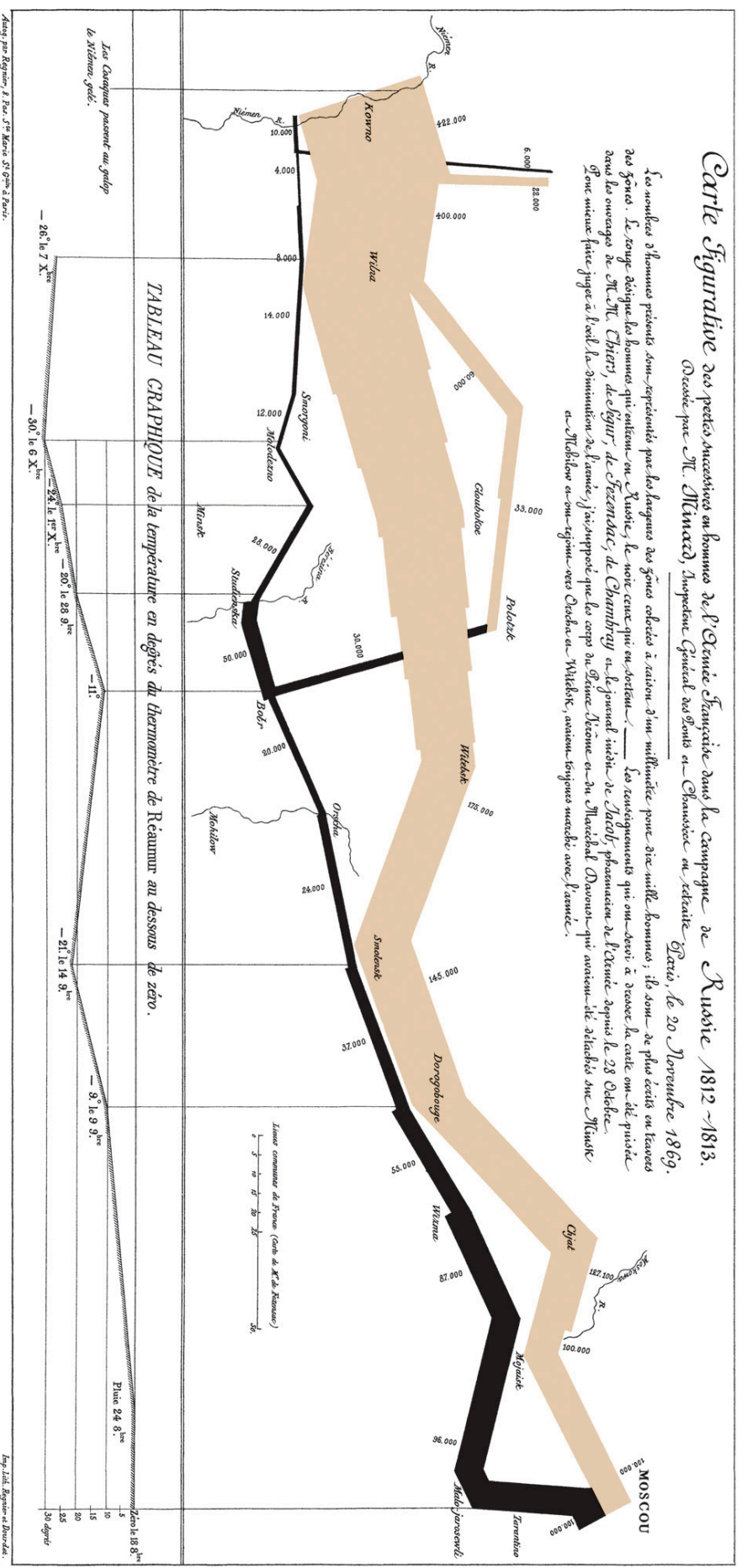


Time Plots

Carte figurative des peccés Maccabéens en hommes de l'Eglise transmise dans la campagne de Russie 1812-1813.

Russie par M. Menard, Inspecteur Général des Douanes et Chambrier en retraite
Paris, le 20 Novembre 1869.

Les membres d'humanité présents sont représentés par les langues des gens collectés à raison d'un millimètre pour dix mille hommes; ils sont de plus écrits en caractères grecs. Le George désigne les hommes qui, antérieurement, ont travaillé, le mot ceux qui ont travaillé. Les renseignements qui ont servi à dresser le carte ont été puisés dans les ouvrages de M. Th. Christ, de Leipzig, de Tcherniak, de Chombarov et le journal russe de Jacoby, pharmacien de Vienne depuis le 28 Octobre. On ne pouvait faire juger à l'œil la diminution des plantes; j'ai rapproché que les corps du genre Nicotia et du Malthoid. D'autant que antérieurement à l'attaque du Malthoid et on trouve ces Chacab et Wilkies, antérieurement avec l'attaque.



+ The plot contains seven variables, each mapped to a different aesthetic:

Information	Aesthetic	Level of Measurement
Size of Napoleon's Grande Armée	Width of path	Ratio
Longitude of the army's position	x-axis	Ratio
Latitude of the army's position	y-axis	Ratio
Direction of the army's movement	Color of path	Nominal
Date of points along retreat path	Text below plot	??????? (What do you think?)
Temperature during the army's retreat	Line below plot	??????? (What do you think?)
Geographic features	Map background	??????? (What do you think?)



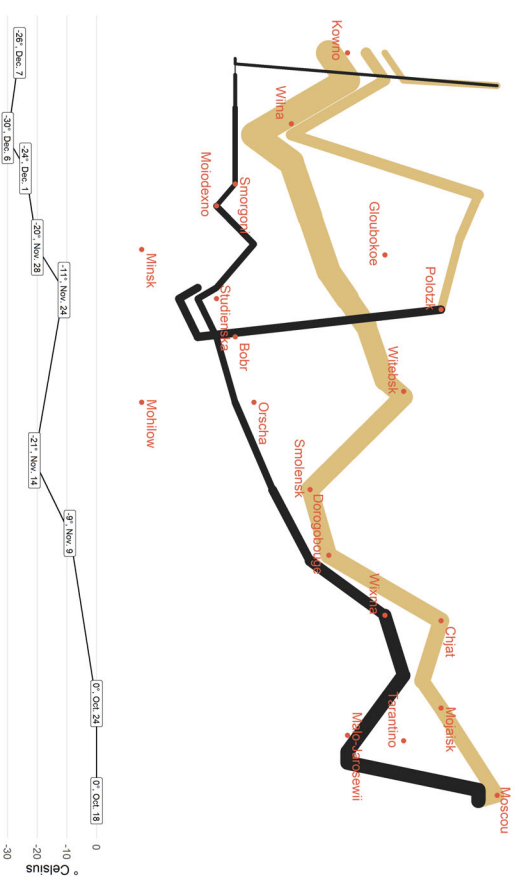
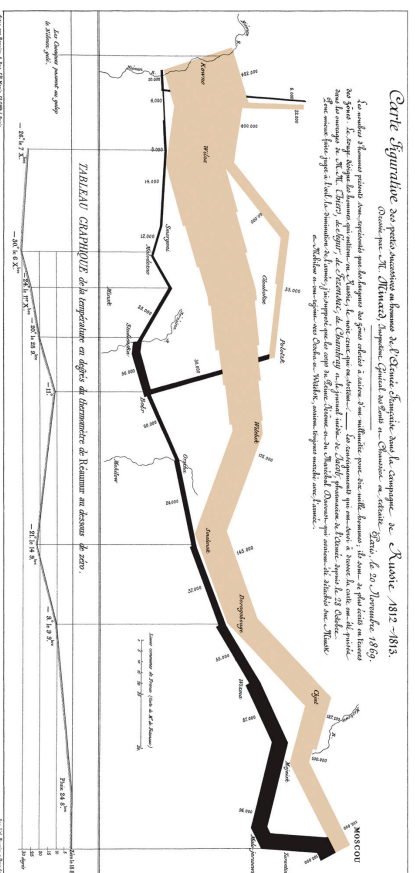
Recreation in R

- + This plot has been recreated in R by:
 - + Andrew Heiss
 - + Michael Friendly
 - + Hadley Wickham



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Side by Side



```
library(tidyverse)
library(lubridate)
library(ggplot2)
library(ggmap)
library(ggrepel)
library(gridExtra)
library(psych)

#Download Directly
download=FALSE # set to true to download
if(download){
  cities <- read.table("https://raw.githubusercontent.com/andrewhelms/fancy-minard/master/input/minard/cities.txt",
                      header = TRUE, stringsAsFactors = FALSE)
} else{
  #load from local
  cities <- read.table("dat/cities.txt",
                      header = TRUE, stringsAsFactors = FALSE)
}

troops <- read.table("https://raw.githubusercontent.com/andrewhelms/fancy-minard/master/input/minard/troops.txt",
                    header = TRUE, stringsAsFactors = FALSE)
temps <- read.table("https://raw.githubusercontent.com/andrewhelms/fancy-minard/master/input/minard/temps.txt",
                    header = TRUE, stringsAsFactors = FALSE)
}
```

```
## vars n mean sd median trimmed mad min max range
## long 1 20 30.79 4.08 30.30 36.77 4.74 24.0 37.6 13.6
## lat 2 20 54.87 0.35 54.95 54.89 0.74 53.9 55.8 1.9
## city* 3 20 10.50 5.92 10.50 10.50 7.41 1.0 20.0 19.0
## skew kurtosis se
## long 0.17 -1.31 0.91
## lat -0.20 -1.18 0.12
## city* 0.00 -1.38 1.32
```

describe(troops)

```
## vars n mean sd median trimmed mad min max range
## long 1 51 28.97 4.39 28.3 28.56 5.49
## lat 2 51 54.93 0.51 54.9 54.91 0.74
## survivors 3 51 91217.65 101718.61 40000.0 72880.49 50408.40
## direction* 4 51 1.51 0.50 2.0 1.51 0.00
## group 5 51 1.43 0.70 1.0 1.29 0.00
## min max range skew kurtosis se
## long 24.0 37.7 13.7 0.64 -0.88 0.61
## lat 54.1 55.8 1.7 0.13 -1.29 0.07
## survivors 4000.0 340000.0 336000.0 1.31 0.48 14243.45
## direction* 1.0 2.0 1.0 -0.04 -2.04 0.07
## group 1.0 3.0 2.0 1.27 0.15 0.10
```

describe(temps)

```
## vars n mean sd median trimmed mad min max range
## long 1 9 30.63 4.30 29.2 30.63 4.15 25.3 37.6 12.3
## temp 2 9 -15.67 11.10 -20.0 -15.67 13.34 -30.0 0.0 30.0
## month* 3 9 1.89 0.78 2.0 1.89 1.48 1.0 3.0 2.0
## day 4 9 14.56 9.46 14.0 14.56 11.86 1.0 28.0 27.0
## date* 5 9 5.00 2.74 5.0 5.00 2.97 1.0 9.0 8.0
## skew kurtosis se
## long -1.56 1.43
## temp -1.66 3.70
```



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More Accessible Resources

- + R Graph Catalog
- + intro
- + Data Science for Psychologists



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Wrapping Up...



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